Exercises

Exercise 9.1 · Computations. Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 4 & -2 \end{pmatrix}$. Compute $A + B$, $2A - B$, AB and BA.

Exercise 9.2 · Inverse and system of order 2. Let $A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$.

1. Explain why A is invertible and find A^{-1}.
2. Deduce the solution of the system $3x + 5y = 4$, $x + 2y = 1$.

Exercise 9.3 · System of order 3. Consider the system $x+y+z = 6$, $2x-y+z = 3$, $x+2y-z = 2$.

1. Write it in the form $AX = B$.
2. With a calculator, find A^{-1}, then the solution of the system.
3. Check the solution by hand.

Exercise 9.4 · Powers by induction. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Compute A^2 and A^3, conjecture an expression of A^n for $n \geq 1$, then prove it by induction.

Exercise 9.5 · A polynomial relation. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

1. Check that $A^2 = 2A + 3I_2$.
2. Deduce that A is invertible and express A^{-1} in terms of A and I_2. Check with the formula of the course.

Exercise 9.6 · Rotations.

1. Using the addition formulas, prove that $R_\theta R_{\theta'} = R_{\theta+\theta'}$.
2. Deduce $\left(R_{\pi/6}\right)^{12}$.
3. Find the image of the point $(2, 0)$ under the rotation with centre O and angle $\frac{\pi}{3}$.

Exercise 9.7 · A parabola through three points. Find the real numbers a, b, c such that the parabola with equation $y = ax^2 + bx + c$ passes through the points $(1, 2)$, $(2, 3)$ and $(3, 6)$. Write the system in matrix form.